

Business Needs generation — EDA

Question: are the generated Business Needs **grounded** (no invention) and **complete**, do they **use every selected stage**, and are each stage's needs **in-scope** for that stage?

What is generated

Business Needs are generated **per approved value stream**, for its **selected stages** (here: the GT stages), from the **raw idea card only** (no summary, no generation signals — the locked theme-gen decision). The output is one consolidated document structured as one block per stage:

```
Value Stage: <stage name>
  Business Product Feature: <scope area>
    1. <business need>  Note / Dependency / Business Rule (each only if stated)
      Operational Training / Operational Reporting  (only if stated)
    <repeat per selected stage>
```

How we evaluate — reference-free + structural

Like description, we do **not** score against the free-form GT text (matching it penalises style). We judge each value stream's Business Needs against the **raw source**, plus two **structural** checks the stage-keyed format makes possible:

metric	measures	vs
faithfulness	claims grounded in the idea card (no invention)	source
hallucination	1 - faithfulness (the invented claims)	source
coverage	the idea card's key facts reflected	source
stage_usage	selected stages addressed in the output	the stage set
stage_align	addressed stages whose needs fit that stage's scope (not misfiled)	catalogue scope

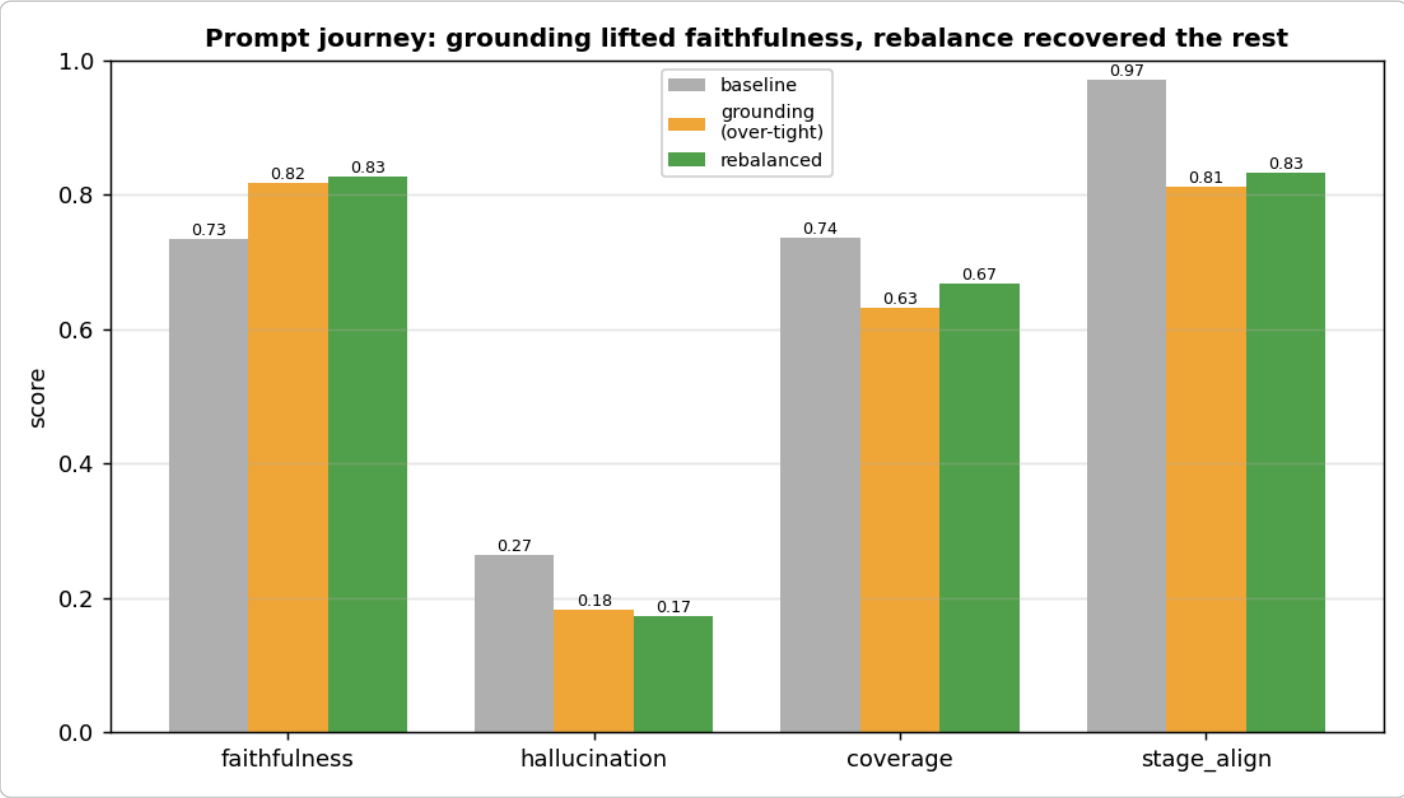
Judge note: the faithfulness/coverage/stage judges run on the **same gateway model that generated the text** (`gpt-5-mini`). A model scores its own output a little favourably, so the absolute numbers may be mildly optimistic; the **deltas** across prompt versions are valid (same judge throughout). A `--judge-model` flag exists to re-judge with a stronger model (e.g. GPT-5).

Finding — the structural checks pass; grounding is the lever

The **stage checks were excellent from the start** and held throughout: the model **uses every selected stage** and files needs under the right one.

metric	baseline	final
stage_usage	1.000	0.998
stage_align	0.972	0.833

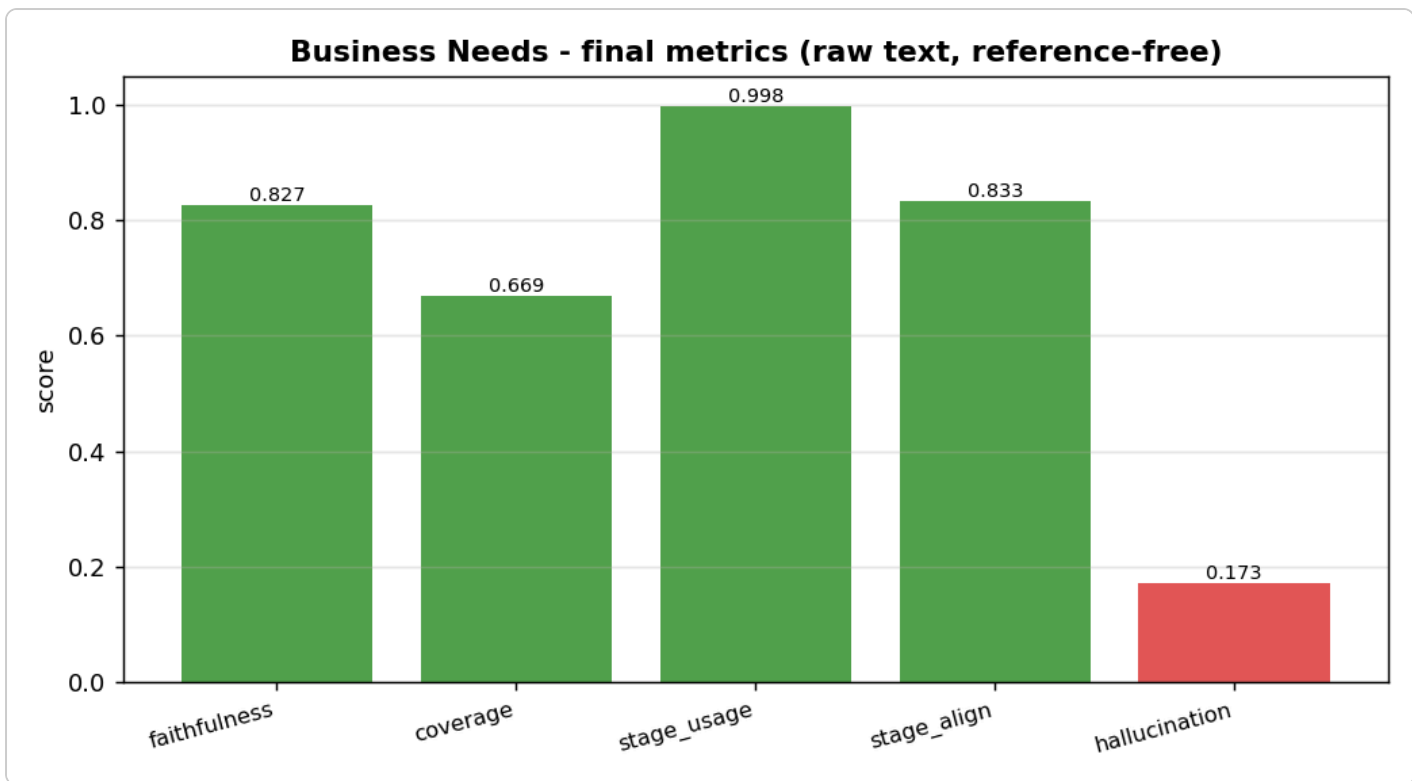
The real work was **grounding**. Business Needs is **prescriptive** ("the business needs X must be built"), so it invents requirements far more readily than descriptive prose — the baseline leaked 26% hallucination.



The journey (same 50-ticket sample, seed 13):

metric	baseline	grounding (over-tight)	rebalanced
faithfulness	0.735	0.817	0.827
hallucination	0.265	0.183	0.173
coverage	0.736	0.633	0.669
stage_align	0.972	0.812	0.833

- 1. Diagnosis:** the prompt was **signal-centric** (Operational Training/Reporting "only if signals exist"; Note/Dependency/Business Rule) — but theme gen feeds **raw text with no signals**, so the model **fabricated** dependencies, business rules, and training/reporting from inference. 1 in 4 claims unsupported.
- 2. Grounding pass:** a hard rule (every need/note/dependency/rule must trace to a card phrase; conditional sub-fields reframed from "signals" to "the card explicitly states it; never infer") lifted faithfulness **0.74 → 0.82**, but **over-corrected**: "a thin stage gets few needs / don't pad" made the per-stage needs **sparse and vague**, dropping coverage (0.74 → 0.63) and stage_align (0.97 → 0.81) — vague one-liners are hard to scope-match.
- 3. Rebalance:** reworted from "write less" to "write only what's grounded, but capture **everything** the card supports and state each need concretely enough to belong to **THAT** stage." This **kept the no-invention gain and recovered alignment** (0.81 → 0.83) and some coverage (0.63 → 0.67), with faithfulness at its best (0.827).



The honest read

- **Faithfulness 0.83 / hallucination 0.17** is solid for a **prescriptive** artifact — but it will not reach description's 0.94. Inventing requirements is the inherent failure mode of "business needs"; ~0.17 is close to its practical floor on raw-only input.
- **Coverage 0.67** is the remaining tension: grounding harder permanently costs some completeness (it recovered only partway from the over-tight pass). This is an accepted trade — **a BA can add a missed need; they cannot trust a fabricated requirement, dependency, or rule**. No-invention is the domain's #1 rule, so we optimise faithfulness over completeness.
- **The structural checks are the win you asked for:** stage_usage $\approx$ 1.0 (no stage silently dropped) and stage_align 0.83 (needs filed under the right stage) — the document is well-structured.

Verdict — locked

Business Needs: raw idea card only, grounding-rebalanced prompt → **faithfulness 0.827 / hallucination 0.173 / coverage 0.669 / stage_usage 0.998 / stage_align 0.833.**

- Reference-free (judged vs source; the GT format is too varied to score against), plus the stage-usage / stage-alignment structural checks.
- Grounding was the lever (hallucination 0.27 → 0.17); the rebalance kept it while recovering stage alignment and some coverage.
- The ~0.17 hallucination is near the practical floor for a prescriptive artifact, and the coverage cost is the accepted no-invention trade.

No further changes (optional: re-judge with GPT-5 via `--judge-model` for a stricter absolute read — the deltas hold either way).